What makes an asset the dominant intermediary?

Centre for Blockchain Technologies

The Making of Dominant Vehicle Currencies

Evidence from DeFi

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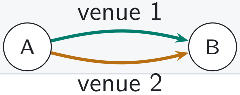
How we reconstruct a route from one transaction

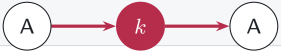
Direct swap
one transaction

Sequential vehicle route
one transaction

Parallel split
one transaction



Round trip
one transaction

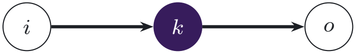


Swap events in one transaction reveal sequential routes, parallel splits, and round trips.

Vehicle dominance is the share of indirect trade carried by an asset

MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset k.
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly k connects endpoint pairs.

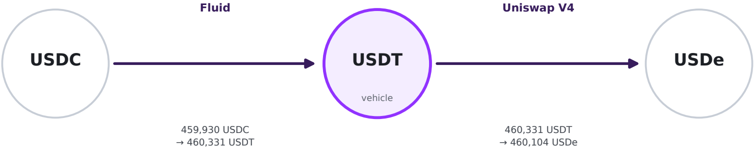


Count-weighted dominance
Share of indirect routes that pass through k

Value-weighted dominance
Share of comparable routed value that passes through k

This transaction routes USDC into USDe through USDT

OBSERVED ROUTE

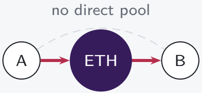


$459,930 observed route value · two venues · one transaction

The trace shows that USDT was used. It cannot tell us whether a direct USDC–USDe route was available or cheaper at that moment.

Architecture changes routing opportunities

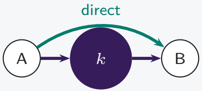
V1 mandatory hub



Only ETH–token pools: token-to-token execution must pass through ETH.

Hub use is imposed

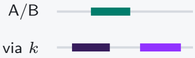
V2 pair choice



Arbitrary pairs add a direct route; vehicle routes remain feasible.

Hub use becomes a choice

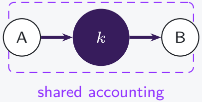
V3 range choice



LPs choose ranges and fees; active depth can favor direct pairs or vehicle spokes.

Opportunity shifts by pair

V4 net settlement



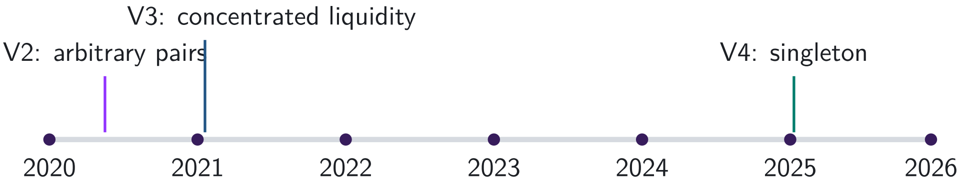
Two pools still execute; intermediate token movement may be netted.

Settlement and vehicle use can diverge

Architecture feasible rules **Exposure** realised pool and route use **Calendar date** timing only

The sample spans Uniswap V1 through V4

DATA



Routes are reconstructed transaction by transaction across the routed venue perimeter. Daily and weekly panels are derived from the same route units.

Vehicle dominance aggregates realised intermediary choices

One realised route

k occupies the intermediary position

Vehicle dominance

D_k = use through k
÷ total intermediary use

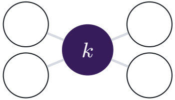
Count or value weighting

Excess vehicle use

$X_k = D_k \div E_k$
E_k: endpoint share of k

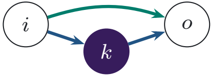
$X_k > 1$: intermediary use exceeds the endpoint benchmark

Separate characteristics of the asset and route set



Network position

position of k in the feasible graph

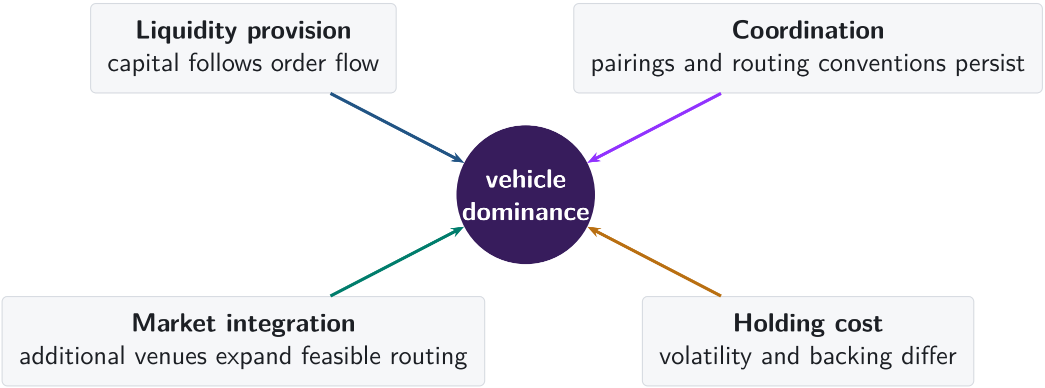


Execution cost

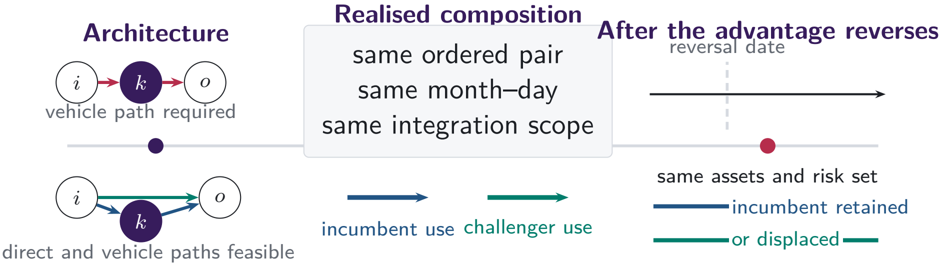
which route returns more at the same state

Endpoint demand supplies the benchmark; network position and same-state route costs remain separate characteristics.

Why might one asset become the dominant vehicle?

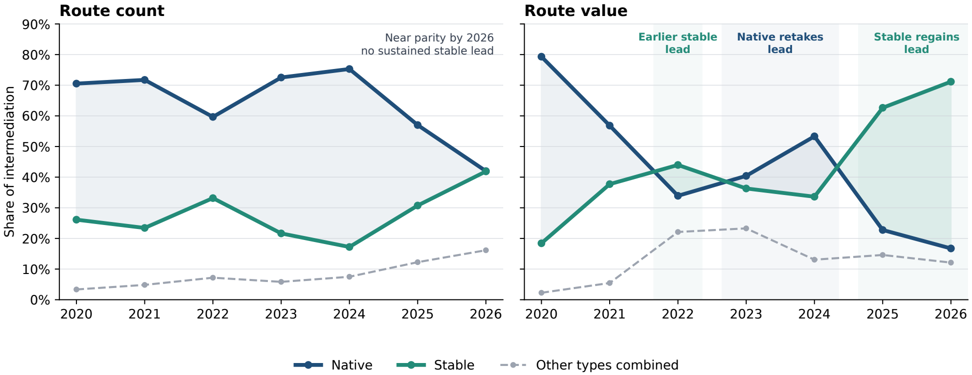


Architecture opens routes; realised trades reveal which vehicle is used



Conceptual design sequence — no estimates shown

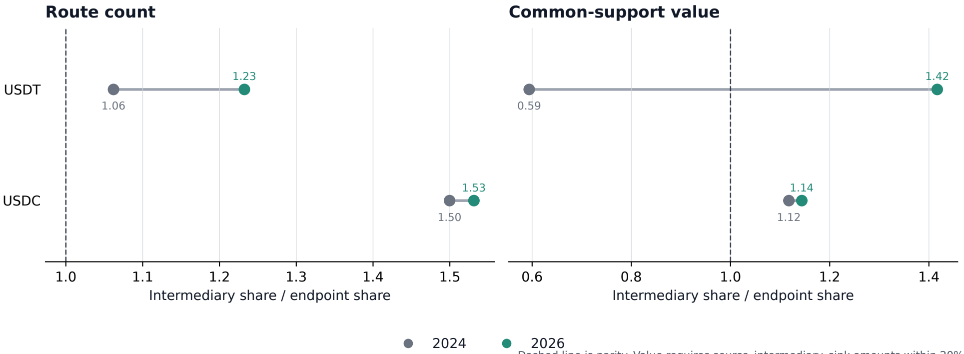
Stablecoins regain the lead in routed value



For value-weighted shares, source, intermediary, and destination dollar amounts must agree within 20%.

Annual shares of observed vehicle routes; the figure does not attribute the changes to protocol design.

USDT drives the post-2024 rise in stablecoin intermediation

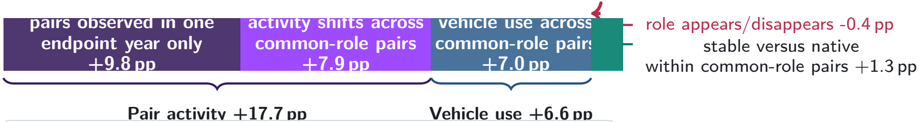


Dashed line is parity. Value requires source–intermediary–sink amounts within 20%.

The value panel retains routes whose source, intermediary, and destination dollar values agree within 20%. Parity means that intermediary and endpoint shares are equal. USDT crosses parity by routed value; USDC remains above it in both years.

Pair activity and vehicle use: 94.9% of the gain

Stable share: 16.9% $\longrightarrow$ 42.5% **+25.7 pp**



Among pairs observed in both endpoint years, stable share still rises
26.1% $\longrightarrow$ 41.9% **+15.8 pp**

Supported value tells the same broad story. Stable share rises +42.8 pp, with +26.2 pp from activity weights among continuing pairs and only 0.0 pp from stable-for-native substitution within them.

Route-count accounting for ordered endpoint pairs, pooling single- and cross-venue routes on month-days observed in both endpoint years through 30 June. Pair activity is eligible activity admitted to the route-component ledger; vehicle use is an exact two-leg route with a native or stable intermediary. The terms describe observed routing, not feasible routes or causal effects.

Cheaper challengers gain share as incumbents lose it

30-day outcome	$\hat{\beta}$	**SE**	t
Challenger vehicle-share response	4.825	0.366	13.17
Incumbent vehicle-share response	-3.872	0.420	-9.22
Challenger overtakes incumbent	9.947	0.486	20.48

$N = 57,989$ pair-candidate observations; 2,213 date clusters. Exact alternating-projection absorption and CR1 rank correction.

The estimates are large in the small set of pair-candidate observations we can follow. They remain descriptive because the panel does not compare each pair against a fixed set of feasible routes.

Current architecture changes do not provide a clean before–after comparison

	5%	10%	25%
Sustained entries	162 (0)	139 (0)	78 (0)
Substitution exits	79 (0)	69 (0)	54 (0)
Role disappearances	7 (3)	7 (3)	7 (3)
Immediate entry contrast	–	–	–
Immediate exit contrast	–	–	–

At every threshold, entries and exits coincide with changes in the comparison set or with the appearance and disappearance of vehicle roles. These windows therefore cannot separate an architecture effect from changes in which pairs and vehicles remain active.

A before–after estimate requires the same feasible alternatives before and after the architecture change.

Stablecoins regain the value lead as route counts approach parity

- Stablecoins lead routed value in 2022 and again in 2025–2026, but fall behind native assets in 2023–2024.
- USDT drives the post-2024 rise in excess stablecoin intermediation; USDC changes little.
- Pair activity and vehicle use account for almost all of the aggregate stablecoin gain. Their economic cause remains open.

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

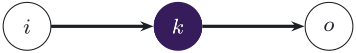
Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Endpoint demand

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

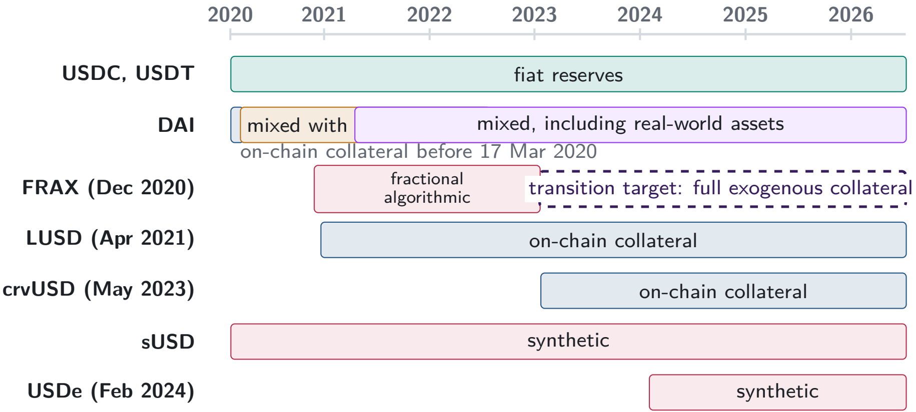
A1. One reconstructed route is one unit



one route unit
two swap legs

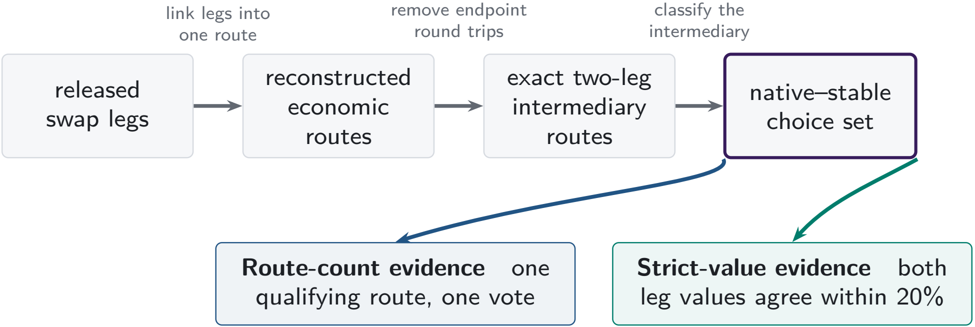
- Each linked $i \to k \to o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k.

A2. Stablecoin backing cannot be treated as fixed



Bars begin at public launch or the sample start for older tokens. DAI cells mark admissible backing; the dashed FRAX arrow is a governance target, not an observed collateral share.

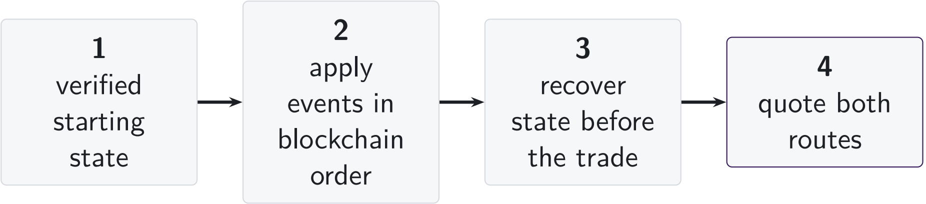
A3. One route universe supports two measurements



Topology-valid routes remain in count evidence when dollar support is unavailable.
Box widths do not represent attrition.

Missing dollar support narrows value-weighted evidence; it does not erase a topology-valid route from the count evidence.

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hooks
Constant product	■	–	–	–	–	■	–
Concentrated liquidity	■	■	–	–	–	■	–
StableSwap families	■	–	■	–	■	■	–
Weighted pools	■	–	–	■	■	■	–
V4 singleton	■	■	–	–	–	■	□

The common object is the state immediately before execution. The fields required to recover that state vary with the pricing rule.

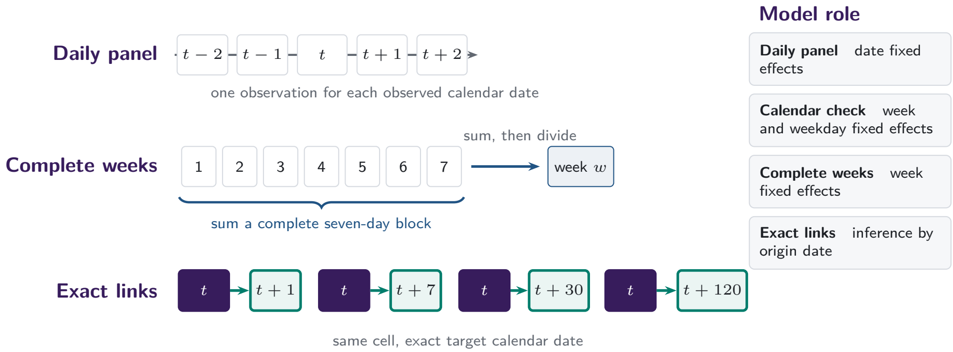
■ required state
□ hooks outside comparison

A6. V1 forced routes have an on-chain signature

- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.

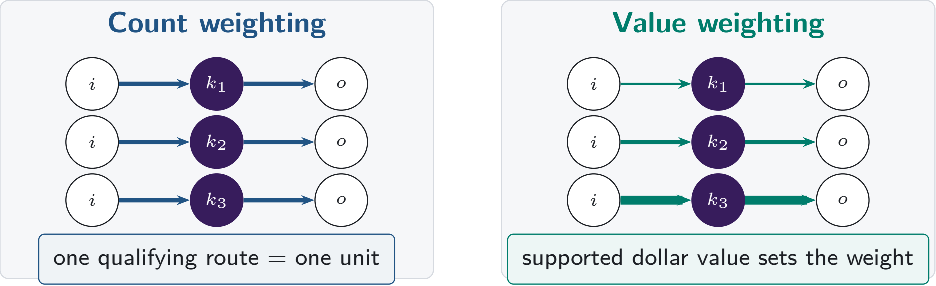
same transaction, matching ETH flow

A7. Daily and weekly frequencies answer different questions



Missing target dates remain missing: row shifts and calendar months do not substitute for exact calendar links.

A8. Count and value define separate dominance margins



Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

A9. Endpoint demand and intermediary use are separate margins

Intermediary share

$$S^I_{k,t} = \frac{N^I_{k,t}}{\sum_j N^I_{j,t}}$$

How often asset k sits inside a route.

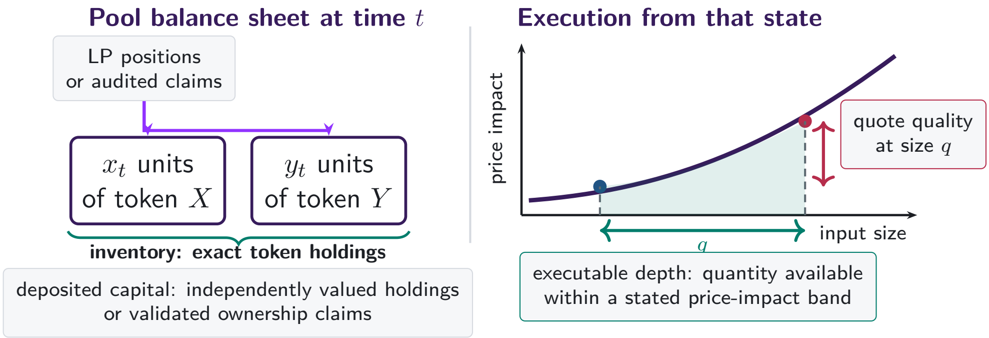
Endpoint share

$$S^E_{k,t} = \frac{N^{\text{in}}_{k,t} + N^{\text{out}}_{k,t}}{\sum_j N^{\text{in}}_{j,t} + N^{\text{out}}_{j,t}}$$

How often asset k is what traders buy or sell.

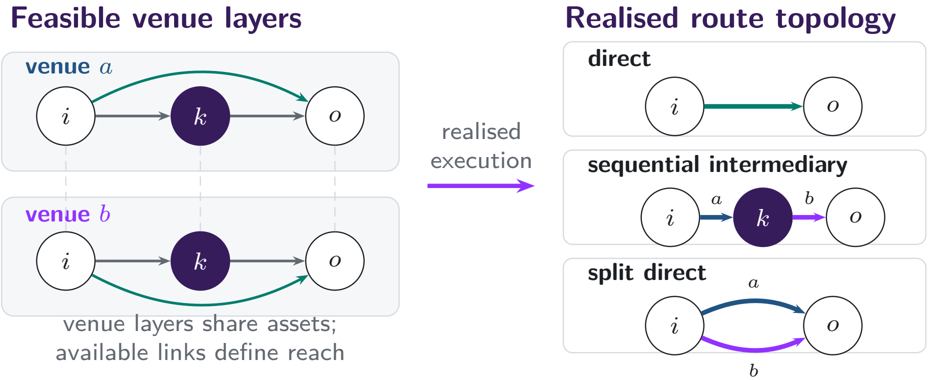
Excess use compares intermediation with endpoint demand on the same support.

A10. Capital, inventory, and depth are different objects



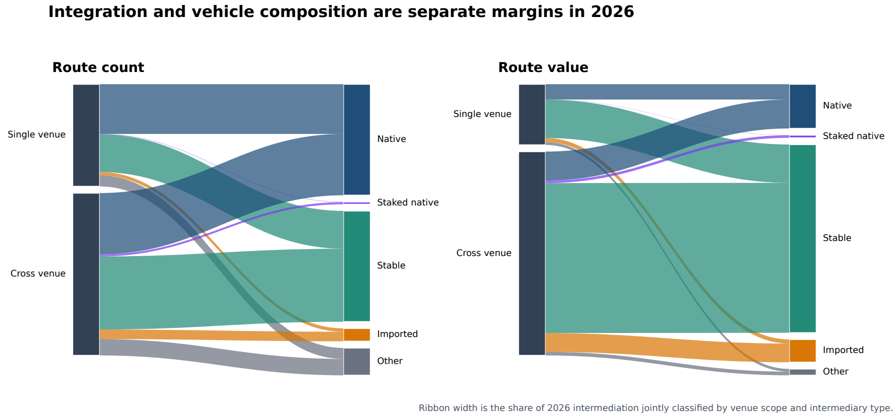
Capital is a stock. Depth and quotes depend on trade size and current pool state.

A11. Integration changes the routes that can be chosen



The analysis therefore separates opportunity-set reach from the composition of realised routes.

A11b. Venue scope and vehicle type differ in 2026



The ribbons classify realised intermediation jointly by venue scope and intermediary type. They describe selected routes; they do not identify an effect of integration or the unchosen opportunity set.

A12. References: vehicle currencies and market structure

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- Dowd and Greenaway (1993), *Economic Journal*
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- Krugman (1980), *Journal of Money, Credit and Banking*
- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: decentralised exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
- Barbon and Ranaldo, decentralised exchange costs and market structure
- Lehar and Parlour (2024), *Journal of Finance*
- Milionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
- Xu, Paruch, Cousaert and Feng, AMM design and slippage